

STEPHEN "CHRIS" HULL

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Software Engineering Leader | Edge & Embedded AI | Distributed Systems | Multi-Cloud Platforms

Engineering leader with 14+ years of experience designing and leading production platforms spanning distributed systems, edge and embedded computing, real-time computer vision, IoT, machine learning, and multi-cloud architecture. Proven track record setting technical direction, leading cross-functional architecture, and delivering complex platforms from embedded hardware through cloud services and ML systems. Deep expertise in NVIDIA Jetson and edge AI, real-time computer vision, Kubernetes, Kafka, gRPC, and cloud-native architectures across AWS and Google Cloud.

EXPERIENCE

ACV Auctions

Buffalo, NY (Remote)

Staff Software Engineer - Research & Development

May 2022 - Present

Viper - Edge & Computer Vision

- Architected and implemented Viper, a production edge-to-cloud vehicle inspection platform integrating NVIDIA Jetson and Raspberry Pi systems with real-time computer vision, AWS IoT Core, Kafka, S3, FastAPI, PostgreSQL, and machine-learning services.
- Directed architecture across embedded, cloud, and ML teams, defining technical strategy for device communication, multi-camera image and video acquisition, event-driven processing, cloud integration, and ML-powered vehicle condition analysis.
- Engineered high-performance NVIDIA Jetson-based computer vision systems supporting multi-camera capture, GPU-accelerated image processing, and production ML inference at the edge.
- Built automated production provisioning tooling for NVIDIA Jetson Orin NX systems, reducing the time required to flash, configure, and validate the seven modules needed for a complete installation to approximately 30 minutes while enabling repeatable workflows across internal teams and third-party contract manufacturers.
- Led technical development of ACV One, integrating mobile and web applications with Viper edge hardware, real-time inspection data, and ML-powered condition analysis to deliver a unified vehicle-inspection workflow.

ML, Multi-Cloud & Platform Architecture

- Defined production multi-cloud architecture across AWS and Google Cloud spanning Kubernetes, IoT, distributed messaging, cloud storage, GPU-accelerated ML infrastructure, and edge-to-cloud systems.
- Architected and deployed a high-performance ML inference platform on GCP GKE using GPU time-sharing, gRPC microservices, Envoy, and Apache Kafka, serving 15+ models and 198K+ monthly requests while reducing annual cloud costs by \$120K+ and response latency by 80%.
- Standardized observability across 30+ production services, establishing reusable logging, metrics, alerting, and Helm-based monitoring patterns while leading migration from Splunk and New Relic to Datadog.
- Established reference architectures and reusable GitHub templates for Python/FastAPI services, ML deployments, and Kafka consumers, standardizing CI/CD, observability, and service conventions across new engineering projects.

Schellman & Company, LLC

Tampa, FL (Remote)

Manager of Application Development

September 2021 - April 2022

- Led application development and platform architecture across application, infrastructure, identity, observability, and data domains, aligning technical strategy with long-term modernization goals.
- Directed technology evaluation and architectural decision-making across engineering initiatives, establishing consistent patterns for application design, cloud infrastructure, identity, observability, and platform delivery.
- Established architecture review, engineering standards, and development practices that improved consistency across software development, QA, and platform delivery.
- Built and mentored a multidisciplinary engineering team, providing technical mentorship, architectural guidance, and development practices that strengthened team capability and long-term system ownership.

Senior Software Engineer

November 2019 - September 2021

- Defined and drove the modernization roadmap for the AuditSource platform, evaluating strategic migrations across storage, identity, observability, and persistence, including Box → S3, Keycloak → Okta, Splunk/New Relic/Sentry → Datadog, and MySQL → Aurora.
- Owned core platform architecture spanning Spring Boot services, Angular applications, enterprise identity, AWS EKS, and production observability, providing technical direction across application and infrastructure concerns.
- Architected enterprise data integrations connecting AuditSource, Quickbase, Workday, and Box with Snowflake and AWS Redshift, establishing a scalable foundation for cross-system reporting and analytics.
- Raised engineering quality standards by introducing automated testing, schema management, and static analysis using Selenium, Robot Framework, Cypress, Liquibase, and SonarQube.

JPMorgan Chase & Co

Tampa, FL

Software Engineer

August 2019 - November 2019

- Developed critical components for a high-volume real-time payments platform using Java, Spring Boot, Apache Kafka, and MySQL, supporting event-driven transaction processing and enterprise-scale financial workloads.
- Evaluated and prototyped cloud-platform modernization through a proof-of-concept migration from Linux VM infrastructure to Pivotal Cloud Foundry.
- Led migration of enterprise Java applications from IBM WebSphere to Linux-based infrastructure, improving deployment flexibility and supporting broader platform modernization initiatives.

General Motors

Austin, TX

Software Development Lead

June 2016 - August 2019

- Led architecture and technical direction for modernization of the vehicle-signing platform, decomposing a monolithic application into scalable Spring Boot microservices using Apache Kafka and PostgreSQL.
- Defined reusable engineering standards across security, automated testing, secrets management, schema management, code quality, and Cloud Foundry deployments, providing architectural guidance across cross-functional teams.
- Led migration from Oracle RDBMS to PostgreSQL and standardized database change management using Liquibase and Jenkins CI/CD, reducing manual deployment dependencies and improving environment consistency and repeatability.

Best Bass Gear, LLC (bestbassgear.com)

Tampa, FL (Remote)

Software Developer

August 2012 - May 2016

- Led migration of the 3dCart eCommerce platform to Magento, redesigning product catalog taxonomy, content structure, and digital assets while preserving 95%+ of existing SEO rankings.
- Developed technical documentation, implementation guides, wiring diagrams, and instructional content while using Google Analytics to measure engagement and improve customer-facing technical resources.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, TypeScript
Architecture & Leadership	Distributed Systems, Cloud-Native Architecture, Microservices Architecture, Event-Driven Architecture, System Design, Software Architecture, Technical Strategy, Architecture Governance, Platform Engineering, Technical Leadership
APIs & Messaging	REST APIs, gRPC, Apache Kafka, RabbitMQ
Cloud & Platform	AWS (EC2, EKS, S3, Lambda, IoT Core, RDS, CloudWatch), Google Cloud Platform (GCP) (GKE, BigQuery, GPU Workloads, NVIDIA Multi-Instance GPU (MIG)), Kubernetes, Docker, Helm, Terraform, Infrastructure as Code, Multi-Cloud Architecture
Edge, Embedded, & Vision	NVIDIA Jetson, Jetson Orin NX, Raspberry Pi, Embedded Linux, Edge Computing, Edge AI, IoT, Device Provisioning, GStreamer, V4L2, CUDA, NVIDIA Multimedia API, Computer Vision, YOLO/YOLOX, ML Inference, GPU Inference, Real-Time Image Processing
Frameworks	Spring, Spring Boot, FastAPI, Django, Angular, React Native
Data	PostgreSQL, MySQL, Redis, Oracle RDBMS, Amazon Aurora, Snowflake, AWS Redshift
DevOps, Observability, & Security	CI/CD, Jenkins, GitHub, GitLab, Datadog, Splunk, New Relic, Sentry, SonarQube, Logging, Metrics, Alerting, HashiCorp Vault, Okta, Keycloak, Secrets Management, Identity & Access Management

PATENTS

- Systems & Techniques for Vehicle Inspection & Condition Analysis**
U.S. Patent No. 12,597,115 | Publications: US-20260063564-A1, US-20260063565-A1

Patented vehicle inspection architecture combining multi-camera exterior and undercarriage imaging, sensor-triggered data capture, computer vision, and machine-learning models to detect vehicle defects and support automated condition analysis.

EDUCATION

University of South Florida	Tampa, Florida
B.S. in Computer Science	- May 2016